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Analysis of electricity user behavior: case study based on results from extended household survey

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Abstract

Greater emphasis on the individual consumers' engagement in energy efficiency has been integrated into the concept of a comprehensive European energy policy. Households, as well as the people living there are very different and one of the important factors is to assess a consumer's behavioral aspects. Conducting surveys is the most common way to obtain this type of information. Our research is based on JSC “Latvenergo” pilot project “Promoting energy efficiency of household using smart technologies” (further – pilot project) where 500 Latvian households were installed with smart meters at the beginning of 2013. The first results have shown that the average electricity consumption in the target group decreased by 23 %. In order to explain the results, research was carried out by analyzing 1) what kind of behavioral and motivation factors influenced changes in electricity consumption; and 2) to what extent changes in electricity consumption were affected by smart meters. Three multiple linear regression models were tested on the basis of first survey data of 430 households and 8 selected households, who were re-interviewed in March – April, 2014. The results show that changes in electricity consumption are not influenced only by technical aspects, but also by users' psychological aspects, such as subconscious and belonging to the social group.

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1. Introduction

Energy consumption is increasing in all sectors caused by the rapid industrial development and population growth. Potential of household energy efficiency improvements is still high. Residential energy consumption

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accounted for 26.7 % of total energy consumption in the EU in 2012 [1]. In 2013 residential buildings consumed 1783 GWh, i.e, 27.1 % of the total final electricity consumption in Latvia [2]. To achieve energy consumption reduction targets in the EU, more and more emphasis is being placed on each consumer involvement in energy efficiency [3, 4].

There are several possibilities to reduce household consumption. In a recent study it was found that standby electricity savings potential of 108 GWh per year can be achieved if Eco-design Directive requirements were met [5]. Another option is through the installation of smart meters. Smart meters provide a dynamic and continuous communication and information flow between energy supply company and end customer, by providing detailed metering information with 5, 10, or 15 minutes recording [6, 7]. The EU has set the ambitious target – Directive 2009/72/EK requires that 80 % of European households must have smart meters installed until 2020 [8]. However, in every country, not just in Europe, there are different aspects of the electricity market and institutions responsible for smart meter implementation. The priority policy objectives and other factors have been implemented increasingly in a variety of smart metering projects. The aim of such programs is to promote the development of smart metering in the future and, consequently, increase energy efficiency in households. A number of smart metering projects have shown that smart meters have great potential to reduce energy use [9–10]. For instance, reduction in electricity consumption by 2.5 % has been achieved by changing user behaviour in the USA [10]. Decrease of electricity consumption by 7.8 % was achieved when educating people about the smart meters and the benefits of data analysis in the Netherlands [11].

It is very important to understand consumers' behavioural aspects, habits of energy use thus allowing to examine various factors that affect electricity consumption. McLoughlin et al. [12] divided the influencing factors into several groups: socio-demographic (income, age, population, time of staying at home, etc.); building physical parameters (household size, number of rooms, age of building, indoor temperature); user behaviour factors (attitude, knowledge, motivation, etc.); electrical equipment (number of equipment, the age of electrical equipment etc.); climatic aspects (location, weather conditions, etc.).

In most European countries installation of smart meters are widely deployed. In Italy and Finland the implementation of smart meters has been made under a strict legal framework. Latvia is still at the early stages of smart metering development compared to other EU countries. However, first steps have been taken. At the end of 2012, JSC "Latvenergo" (the largest electricity supply company in the residential electricity market in Latvia) with co-financing of Climate Change Financial Mechanism have launched the first pilot project. Alongside with the aim of reduction of electricity consumption by 10 %, information on actual electricity consumption and carbon dioxide emissions in order to increase awareness of smart technologies and to promote active participation of households in energy efficiency are other important objectives of the project. In this study data regression analysis was carried out to assess to what extent different factors influence changes in consumption.

2. Materials and methods

2.1. Survey

Household surveys have been identified as the most common method for energy consumption analysis [13]. Before the start of the pilot project, a large household survey was carried out in March – April, 2013. During the survey, important quantitative and qualitative data from households participating in the project were identified by questionnaires: households' users personal, socio-economic, socio-demographic characteristics, number and type of electrical appliances, as well as issues related to attitude, awareness, level of knowledge on energy efficiency, appliance usage habits, behaviour, etc. All households were surveyed through telephone interviews.

For the purpose of this research, face to face interviews with 8 selected households were carried out in March – April, 2014 (i.e. re-survey). Re-survey makes it possible to identify not only the factors that influenced households' electricity consumption, but also helps to identify the participants' attitude, knowledge and behavior aspects in detail within this pilot project. Re-survey was structured in two parts: 1) to ask questions regarding how exactly electricity was used in households and what factors affected different patterns of use in similar households and 2) to estimate

participants' behavioral, motivation and attitude aspects in the context of consumption reduction. Due to protection of personal data, it was not possible to increase the number of potential participants in the re-survey. 8 households were selected based on their own interest to be enrolled in the study.

2.2. Regression analysis

Multiple regression as a data analysis tool has been used widely in recent literature. McLoughlin [12] summarized the most frequently mentioned factors affecting the household's energy consumption: household's income, number of electrical devices, number of household members and type of house. Mondol et al. [14] argued that building location, type of heating, efficiency of electrical equipment and users' attitudes are among the most important. In other studies household's members, the age of the main breadwinner in the household and time of staying home were found as the most statistically significant factors [12, 15–18]. For instance, Zhou and Teng [16] study shows that electricity consumption increased by 8 % when the number of family member increases by one person. Yohanis [17] showed that people who are over 65 consume electricity more than younger people. Some studies have proved that the age of the main breadwinner in the household was positively related to in-home energy consumption levels (instead of average age of all household members) [12, 15–16, 18–20]. Bedir [21] results showed low relationship between the number of electrical equipment and energy consumption ($R^2 = 0.47$). In another studies it was found that building area has a significant impact on electricity consumption [12, 16–17, 22–26]. Bartusch et al. [25] proved that there are not statistically significant variances in annual electricity consumption in respect of living space.

Another interesting study was made by Olaniyen and Evans [26] and Orlando [27] found that price of electricity, household income, climate conditions and technical progress were among factors which significantly affect electricity consumption per capita in month. Based on the literature review, we use some of the factors mostly used in order to elaborate the multiple regression models for our analysis. The proposed regression model is based on estimation of 5 selected independent variables [12, 14–21, 25–28]: total household monthly income (Income, EUR/month); the age of the main breadwinner in the household and the most commonly, the age of the male in the household or husband (Age, years), the number of electrical appliances (Electr. apl.), the number of household members (No of memb.) and the average time of staying at home per month (Time, hours). Similar to Olaniyen and Evans [26] study electricity consumption per capita in month was used as a dependent value in our model. The model is described by the following equation (1):

$$El = \alpha_0 + \alpha_1 \cdot Income + \alpha_2 \cdot Age + \alpha_3 \cdot Electr. apl. + \alpha_4 \cdot No of memb. + \alpha_5 \cdot Time + \varepsilon \quad (1)$$

where

El – electricity consumption per capita in the month, kWh/people month;

α_0 – constant (regression intercept);

$\alpha_1 \dots \alpha_5$ – independent variables explaining deviation of dependent variable El.;

ε – the variance of error, which describes the differences between the variables.

We did not include household area in the model since household area did not change during the first project year in none of the 8 households. Moreover, household area did not show statistically significant correlation with electricity consumption for the target group. It can also be justified based on similar findings from previous studies [25]. Also climatic conditions in relation to the geographical location of the building and the outdoor temperature were not included for regression analysis.

3. Results

3.1. Evaluation of energy efficiency

In 8 households that were interviewed repeatedly, consumption fell by 13 % in 2013, if compared with 2012. Table 1 describes the main characteristics of these households obtained during the re-survey.

Table 1. Main characteristics of 8 households.

Nr	Electricity consumption. kWh/month		Percentage of reduction, %	Type of building	No of memb.	Type of heating	Electr. Apl., number	Age, Years	Income, EUR/month	Time, hours
	2012	2013								
1	641.7	400.1	37.7 %	Private house	4	Pellet boiler	34	35	2000	480
2	568.3	554.3	2.5 %	Private house	3	„Jotul” mantel	31	40	6000	469.5
3	607.3	571.1	6 %	Private house	4	Heat pump	29	41	1200	529.5
4	281.8	205.3	27.1 %	Flat	3	Central heating	37	40	2000	424.5
5	1547.9	1326	15.8 %	Private house	2	Heat pump	43	44	4200	408.0
6	791.9	554.7	29.9 %	Private house	5	Heat pump	30	44	3000	487.5
7	773.3	686.8	11.2 %	Flat	3	Central heating	39	47	1800	480.0
8	100	129	-29 %	Flat	2	Central heating	18	38	1000	540.0

As it can be seen in Table 1, households differ from each other with various characteristics. The highest reduction in electricity consumption was achieved in private houses and which use heat pumps for heating purposes. A similar conclusion can be found in previous studies [29, 30]. It can be explained by the fact that building area and number of electrical appliances in private homes are higher than in apartments. Small changes regarding energy efficiency measures occurred in the first year. Replacements of incandescent bulbs to energy-efficient lights were done in 5 households. The total number of electrical appliances was changed in 3 households. Other energy efficiency actions during the first year have not been implemented. Mostly people are willing to undertake energy efficiency actions that require small financial resources, such as: more efficient lightning, replacement or installation of motion sensors in lighting systems, etc. People are not willing to invest large financial resources in energy efficiency. Users' attitude and behaviour regarding energy efficiency issues asked in re-survey were quite different. Some respondents "saved money from saved kilowatt-hours" mentioned as a main motivation to undertake energy-savings actions. Second important factor was "spirit of competition, where in case of victory households will get some prize. Competition would create gaming and martial vigour". But the normative targets and people's perceptions to become as "green thinking" were listed as insignificant factors.

The awareness level of all respondents is high enough. All respondents mentioned that installation of efficient lighting, disabling of "stand – by" mode, replacement of old appliances and building insulation are the most effective measures to reduce consumption. Most of respondents are willing to reduce electricity consumption only driven by economic incentives, rather than taking care of environment. Consumers do not consider that if they reduce their consumption they become to be "green thinking". At the same time respondents are willing to uptake energy efficiency improvements, but first they will assess economic benefits of these measures and pay-back time. It was mentioned as the most important factor to start energy saving actions. Peoples' opinions on using renewable energy sources for electricity generation are quite positive. However, they would not be ready to invest in new technologies and to pay more for electricity if it is produced in a "greener" manner. People's attitude regarding smart meters is positive, but the interest is falling with time. Respondents confirmed that at the project's early stage they looked into smart meters more often. They were interested to follow their electricity consumption. But after a longer period of time, interest decreased and was even lost. It can be concluded that it does not matter whether smart

meter is installed in the household, because most of the households do not use this opportunity to see detailed consumption data in the Internet.

3.2. Results of regression analysis

A multivariate regression analysis software package, STATGRAPHICS Centurion XVI (version 2.16.04.) is used to estimate the coefficients ($\alpha_0 \dots \alpha_5$) associated with each variable shown in equation (1) and to test their significance. 3 linear regression models were analysed and tested:

1. 1st model: all target group households' data from April 1st, 2012 to March 31st, 2013;
2. 2nd model: 8 target group households' data from April 1st, 2012 to March 31st, 2013;
3. 3rd model: 8 target group households' data from April 1st, 2013 to March 31st, 2014.

28 % of respondents did not reply to the question about their total income per month in the first survey, therefore such cases were excluded from further analysis. The average time of staying at home in households was calculated as the average value of time of staying at home on working days and weekends of all households' members. The data set of 342 numbers of cases was included for the 1st model. The data sets of 8 numbers of cases were included for the 2nd and 3rd model.

The multiple regression results of all models are shown in Figure 1. Table 2 summarises the regression equations of empirical relationship among variables, R^2 , t-statistics, p-values and F-ratios. The 1st model (see Fig. 1. a) showed quite low correlation coefficient ($R^2 = 0.274$) indicating that the model explains only 27.4 % of the variability in consumption by selected 5 predictors. Only 3 independent variables showed statistically significant impact on changes in electricity consumption: time of staying at home, number of appliances and income, while household members and age did not show statistically significant correlations. This is according to the findings of previous studies, where time of staying home (as in [12, 17–19]), the number of electrical equipment (as in [20]) and income (as in [25]) were identified among the most statistically significant factors.

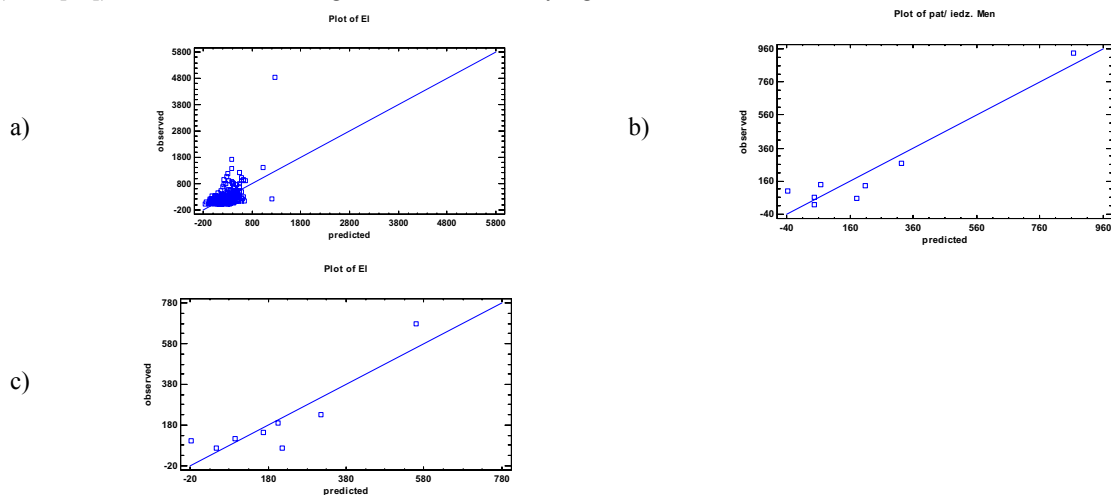


Fig. 1. Multiple regression results: a) 1st model; b) 2nd model; c) 3rd model.

The most part of variance in electricity savings can be explained by other important variables while not included in this model. Since the p-value of the whole model is less than 0.05, it can be concluded that the *model* has no statistically significant predictive capability between the variables at the 95.0 % confidence level. But if we look to

F-test of the 1st model (explains regression is big enough. considering the number of variables needed to achieve it), it can be concluded that F-ratio is quite large indicating that null hypothesis can be rejected.

The results of the 2nd and 3rd model (Fig. 1 b and c) showed that models explain 91.44 % (Adjusted $R^2 = 70.0299$) and 79.67 % (Adjusted $R^2 = 28.8053$) of the variability in consumption, respectively. There is high R^2 value for the 3rd model, but a low adjusted R^2 . It indicates that, probably, we include too many predictors. However p-values of all independent variables included in the 2nd and 3rd model are higher than 0.05 indicating that they are not statistically significant at the 95.0 % or higher confidence level. Both *models* have no statistically significant predictive capability. Also F-ratios for both models are about 5 times lower than in case of the 1st model. But still, models keep the evidence against the null hypothesis, because F-ratios for both models are higher than 1.

Table 2. Linear regression results and overall fit statistics for 3 models tested.

Model N ^o	Regression equation	Parameter	Error	T-statistic	P-value	F-ratio of the model	p-value of the model
1 st model	$El = -20.0781 - 86.5038 \cdot \text{No of memb.} + 0.1042 \cdot \text{Time} + 3.4884 \cdot \text{Age} + 14.5152 \cdot \text{Electr. apl.} - 0.09129 \cdot \text{Income};$ $R^2 = 0.274$; adjusted $R^2 = 0.2376$	CONSTANT	145.087	-0.1384	0.8900	25.36	0.0000
		No of memb.	10.511	9.6385	0.6771		
		Time	0.2213	-0.4168	0.0000		
		Age	1.6038	0.4711	0.6379		
		Electr. apl.	1.5059	-8.2296	0.0000		
		Income	0.2191	2.1750	0.0303		
2 nd model	$El = -1400.43 - 256.375 \cdot \text{No of memb.} + 0.69915 \cdot \text{Time} + 44.2122 \cdot \text{Age} + 18.6239 \cdot \text{Electr. apl.} - 0.206555 \cdot \text{Income};$ $R^2 = 91.4371$; Adjusted $R^2 = 70.0299$	CONSTANT	1804.5	-0.776	0.5189	4.27	0.2005
		No of memb.	74.958	-3.4202	0.0759		
		Time	2.6374	0.2651	0.8158		
		Age	18.1117	2.4412	0.1347		
		Electr. apl.	15.1497	1.2293	0.3439		
		Income	0.1302	-1.5869	0.2534		
3 rd model	$El = -1983.3 - 164.241 \cdot \text{No of memb.} + 2.18777 \cdot \text{Time} + 27.7102 \cdot \text{Age} + 17.7275 \cdot \text{Electr. apl.} - 0.00216664 \cdot \text{Income};$ $R^2 = 79.6587$; Adjusted $R^2 = 28.8053$	CONSTANT	2133.34	-0.9297	0.4507	1.57	0.4337
		No of memb.	87.8891	-1.8687	0.2026		
		Time	3.13994	0.69676	0.5580		
		Age	19.9216	1.39096	0.2988		
		Electr. apl.	16.6872	1.06234	0.3994		
		Income	0.128915	-0.01681	0.9881		

In general, we can conclude that 5 variables included in our analysis are not the most suitable allowing to explain changes in consumption. Most likely, such low p-values and R^2 values in many cases can be explained by the non-inclusion of user behavioural aspects. Probably higher number of cases or precise information from households would better explain the results. From our analysis it is clear, although variables included in the regression analysis showed high R^2 values in the 2nd and 3rd model, variables included in the analysis are not statistically significant. Other factors can be defined and tested in order to improve the results.

3.3. Interpretation of the results

Significant changes in 8 repeatedly interviewed households' had not been occurred during the first year of the pilot project. There were no significant improvements of energy efficiency, and also number of equipment, personal and economic characteristics, as well as their behaviour with regard to the use of electricity has not changed.

Although almost all 8 households decreased consumption after the first year. Multiple regression analysis did not show highly statistically significant correlations among different selected factors. Therefore the explanation why these households achieved so high electricity consumption reduction must be based on more comprehensive analysis based on larger number of cases over a longer period of time. Perhaps, human behaviour is not consistent with words. This means that people conserve energy, but do not recognize it in the interviews.

One explanation could be related to the human subconscious. Smart meter is installed in household, therefore the focus on consumption figures for the smart meter screen happens more often than they can feel it. On a daily basis or a few times a week people alongside the smart meter can see how consumption is increasing and how the possible appliances are being operated. Carrying out activities related to the increase in electricity consumption, people unconsciously think of smart metering. As a result people try to change their behaviour. Another reason could be that respondents in the first months of the pilot project have already recognized which devices consume electricity more. Therefore, they devote more attention to some particular devices they are interested in. Month by month they assess the usefulness of the device in order to save energy and only then realize what they have done. Most of the respondents were born in the Soviet time, when life was relatively modest. Nowadays in the modern, misunderstood democracy forces people to live better, more relaxed and more luxurious. Probably that's why they do not recognize that they could save energy. Recognizing that the respondent saves electricity others may believe that their quality of life, unfortunately, is limited, so they do not meet the standards of society imagined.

4. Conclusions and discussion

Multiple regression models were developed in order to investigate what factors and to what extent can explain variance in electricity consumption. The 1st model showed that Time, Apl, and Income have statistically significant impact on changes in electricity consumption. While No of memb. and AgHoH did not showed statistically significant correlations. Whole model explains only 27.4 % of the variability in consumption. The results of the 2nd and 3rd model did not show statistically significant relationship among the variables and electricity consumption, because p-values of all independent variables were higher than 0.05. The number of household members showed the best p-values in the 2nd and in the 3rd models. The 2nd and 3rd models explain 91.44 % and 79.67 % of the variability in consumption, respectively. In order to provide more qualitative and reliable results, a larger number of cases are needed, as well as analysis over a longer period is preferable.

Re-survey of 8 selected households was carried out with the purpose of clarifying behavioural factors that might influence changes in electricity consumption. Although respondents said they have not introduced much saving measures, reduction of electricity consumption is high (13 %). That means electricity in households has been saved, but people do not feel the saving actions or do not want to recognize that they might, actually, have taken some measures. The main factors for motivation to undertake energy-savings actions are "saved money", "spirit of competition" and "information on real examples how to save". People's attitude regarding smart meters is positive, but it is not the main motivator to decrease their electricity consumption.

Nowadays, the operation of various systems and infrastructure is based on the use of electricity. Therefore, it also suggests that our life would not be unimaginable without electricity. Unfortunately in practice even very high efficiency of electrical equipment does not guarantee expected savings. It is due to the consumers' behavioural aspects. Often people do not know or do not want to change anything in their daily routine. Inefficient use of electricity probably will lead to even more inevitable consequences in the future, therefore increased attention should be paid how to find new innovative solutions for increasing costumers motivation. Smart meters can play a vital role on promoting energy efficiency in households. They are contributing not only to optimize the operation of electricity systems, but also leading to socio-economic platform for the promotion interaction between supplier and consumer.

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